Beyond Nominal Ratios: Language-Aware Prompt Compression for Multi-hop Question Answering

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Abstract. Prompt compression is reported as a ratio: the fraction of the context a compressor keeps. We show that this number is not comparable across languages, and that the gap is large enough to change which method looks cheaper. Measuring token fertility on parallel WMT24++ sentences across four writing systems, a context costs 4.97× more tokens in Hindi, 3.89× in Thai and 2.49× in Vietnamese than its English counterpart under the tokeniser that bills, so a compressor reporting the same nominal ratio in two languages is not delivering the same saving. We propose an *effective ratio* that divides fertility out, and show it separates tokenisers that nominal ratios rank as identical (1.02× spread nominal, 1.77× effective). To ground this we reimplement IterCOMP, a state-of-the-art training-free compressor evaluated only on English, and give it its first systematic evaluation on Vietnamese using the complete VimQA benchmark alongside MuSiQue, 2WikiMultiHopQA and HotpotQA ($n \geq 1{,}003$ each, six compression baselines). Two further effects appear only outside English: a boolean-answer normalisation step is worth +10.0 F1 in Vietnamese and exactly 0.0 in English, and on undiacriticised text – ubiquitous online – compressed context outscores even gold context. We release the implementation and the Vietnamese prompts.

Keywords: Prompt compression · Multi-hop question answering · Tokeniser fertility · Vietnamese NLP · Retrieval-augmented generation

1 Introduction

Multi-hop question answering requires aggregating evidence across documents. Retrieval-augmented generation (RAG) [6] supplies external documents to compensate for a model's bounded knowledge, but introduces two problems: *cost* (input length grows linearly with the number of documents) and *noise* (irrelevant passages degrade reader accuracy).

[0] Corresponding author: Thanh-Nhan Truong, 25C1505320@student.hcmus.edu.vn. Code, prompts (Vietnamese and English), per-dataset results and the extended analyses are at https://github.com/nhantrnh/ANLP-IterCOMP.

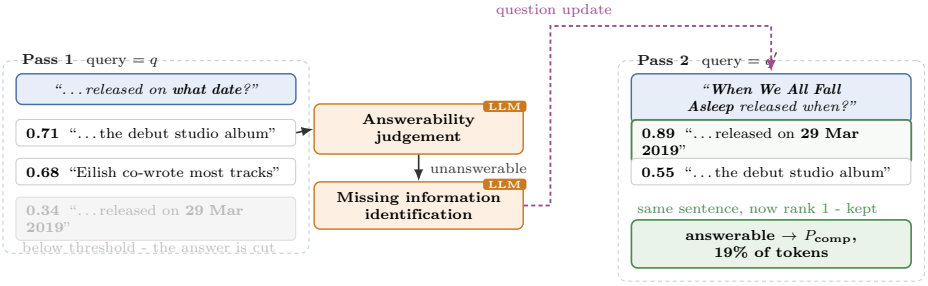


Fig. 1. One VimQA question through two IterCOMP passes. The sentence carrying the release date scores 0.34 and is filtered out, so a single-pass compressor loses the answer; re-scored against the follow-up q' it rises to 0.89 and is kept.

Prompt compression addresses both. However, existing methods [3,9,13,7] compress *once*, conditioned on the original question, and therefore miss the defining property of multi-hop QA: the information needed at a later hop only becomes identifiable after the evidence of the earlier hop is in hand. IterCOMP [14] embeds multi-hop reasoning into the compression loop itself. No public implementation accompanies the original paper, and its evaluation remains restricted to English.

The Vietnamese adaptation and evaluation are our primary contribution; the reproduction that enables them is reported subsequently. Our study contributes five findings, four of them specific to Vietnamese: (1) a boolean-answer normalisation step worth +10.0 F1 on Vietnamese ($p = 0.01$) and 0.0 on English ($p = 1.00$), a linguistic fix rather than a metric-inflating one; (2) evidence that the iteration budget saturates beyond two passes, with `max_iter` from 2 to 5 moving F1 by +1.2 at $n = 500$ (within noise) while retention grows from 0.160 to 0.194; (3) an *effective ratio* showing that nominal compression ratios understate real token cost on Vietnamese by 2.39×, since the tokeniser that bills is not the one that counts; (4) that Vietnamese does not handicap the compressor, IterCOMP reaching 84% of the Oracle ceiling on VimQA, the lower 54% on English MuSiQue being a dataset property, not a language effect (Section 7.4); and (5) that on undiacriticised Vietnamese, compressed context beats gold context by 3.78 to 5.82 F1 (Table 7). We call the resulting configuration IterCOMP-X, and stress that it is *not* a new algorithm - the loop is unchanged; it is a deployment configuration plus a metric-level fix, and we report below which parts survive significance testing.

None of this is measurable without a working reimplementation; absent a public one, we rebuild it from the paper's description and redesign every prompt. It recovers the central result on all four datasets (Table 5); Section 7.2 compares our numbers with the original's, and the released repository carries the per-dataset breakdown.

2 Nominal Ratios Are Not Comparable Across Languages

A compression ratio is reported as a fraction of the context kept, and read as a saving. That reading assumes the unit being counted costs the same everywhere. It does not: the tokeniser that bills splits some writing systems far more finely than English, so the same nominal ratio buys different savings in different languages. This section measures the gap and proposes a correction.

Tokeniser	Vi tok	En tok	Fertility
`cl100k_base`[a]	105.6	44.1	**2.39**×
`o200k_base`[b]	63.3	43.3	1.49×
XLM-R (used by bge-m3)	57.9	49.7	1.19×
`\w+` (unit of "ratio")	47.6	35.7	1.34×

Table 1. Token fertility over 930 parallel English-Vietnamese sentence pairs from WMT24++ [2]. All 95% confidence intervals are narrower than ±0.04. Fertility depends strongly on the tokeniser: a factor of two separates `cl100k_base` from XLM-R. [a]GPT-4, GPT-3.5-turbo; [b]GPT-4o. Which tokeniser bills depends on the served model, so exposure to fertility is set by that choice as much as by the language.

2.1 Four writing systems

The claim above is about Vietnamese, so it needs a control: if every language were billed alike, fertility would not be worth reporting. Measuring 400 parallel WMT24++ pairs per language (`fertility_multilingual.py`) shows it is not.

Language	`cl100k_base`	`o200k_base`
Hindi	**4.97**×	1.71×
Thai	**3.89**×	1.83×
Vietnamese	**2.49**×	1.52×
Chinese	**1.87**×	1.27×

Table 2. Fertility against English over 400 WMT24++ pairs per language, one writing system each: Devanagari, Thai (unspaced), Latin-plus-tones, Han. All 95% confidence intervals are non-overlapping across languages.

Every language exceeds 1.0 with non-overlapping intervals (Table 2), so the mismatch is systematic rather than a Vietnamese quirk; Vietnamese at 2.49× is the *mild* case, Hindi being penalised twice as heavily. The 2.49× here (400 pairs) and the 2.39× in Table 1 (930 pairs) are different samples of the same corpus.

Crucially, fertility depends on the tokeniser measured. The compression ratio (Table 6) counts with `\w+`, where fertility is only 1.34×, so comparing ratios

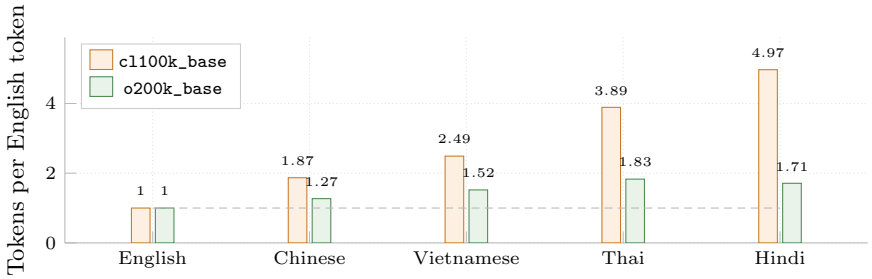


Fig. 2. What the same content costs in tokens, relative to its English translation, over 400 WMT24++ pairs per language. English is 1.0 by construction (dashed). Under `cl100k_base` a Hindi context costs almost five times its English counterpart, so two systems reporting the same nominal ratio in Hindi and in English are not delivering the same saving. Changing the tokeniser changes the tax more than changing the language does: `o200k_base` flattens every bar, and reorders Thai above Hindi.

across languages is close to fair; but *API cost* is billed on the served model's tokeniser - 2.39× under `cl100k_base` against 1.49× under `o200k_base` - while bge-m3's XLM-R is only 1.19×. On the 200 VimQA contexts here a context averages 1,110 `\w+` words but 2,364 `cl100k_base` tokens, so our ratios *understate* billing 2×.

2.2 The effective ratio

Write $|x|_{\mathcal{T}}$ for the number of tokens the billed tokeniser $\mathcal{T}$ assigns to a string x, and let D be the retrieved documents and P_{comp} the compressed prompt. The quantity compression papers report is

$$\text{Ratio}_{\text{nom}} = \frac{|P_{\text{comp}}|_{\mathcal{T}}}{|D|_{\mathcal{T}}}. \tag{1}$$

Let $\phi_{\mathcal{T}}(\mathcal{L})$ be the fertility of language $\mathcal{L}$ under $\mathcal{T}$: the mean token count of a text in $\mathcal{L}$ divided by that of its English translation, which is what Table 2 measures. We define the *effective ratio* as the nominal ratio expressed in English-equivalent tokens,

$$\text{Ratio}_{\text{eff}} = \frac{\text{Ratio}_{\text{nom}}}{\phi_{\mathcal{T}}(\mathcal{L})}, \tag{2}$$

so that $\text{Ratio}_{\text{eff}} = \text{Ratio}_{\text{nom}}$ for English by construction, and two systems are comparable only once both are read on this scale. The correction is deliberately a single division: fertility is a property of the language-tokeniser pair, not of the compressor, so it should be divided out rather than folded into the compressor's score. Reporting the pair $(\text{Ratio}_{\text{nom}}, \text{Ratio}_{\text{eff}})$ costs one extra column and keeps the familiar number intact.

2.3 Does the effective ratio discriminate?

A second metric that moved with the first would add nothing. Holding the compressed contexts fixed - the same 50 VimQA questions, the same IterCOMP output, only the tokeniser changed - the nominal ratio barely moves while the effective ratio does:

Tokeniser	nominal	effective
`cl100k_base`	0.236	**0.099**
`o200k_base`	0.240	0.161
`\w+`	0.234	**0.175**
spread	1.02×	**1.77**×

Because the contexts are byte-identical across the three rows, the whole of that 1.77× is the tokeniser's and none of it the compressor's. A nominal ratio reported on its own therefore conceals all of it, which is the case for adding the second column rather than replacing the first.

3 Related Work

Prompt compression. Extractive methods retain a subset of tokens or sentences: LLMLingua [3] uses perplexity from a small model to drop uninformative tokens; LongLLMLingua [4] adds query awareness; LLMLingua-2 [9] recasts the task as token classification, which is far faster and *multilingual* - the decisive property for our Vietnamese experiments; Selective-Context [7] removes redundant lexical units. *Abstractive* methods regenerate shorter content, notably RECOMP [13]. All compress in a single pass.

Iterative retrieval. Self-Ask [11] and IRCoT [12] decompose questions and interleave retrieval with reasoning; IterCOMP instead *narrows* the context rather than expanding it.

Vietnamese NLP. Foundation models include PhoBERT [8] and ViT5 [10]; to our knowledge, none evaluates prompt compression for Vietnamese.

4 Dataset

The three English datasets are standard for multi-hop QA and are exactly those used by the original paper. VimQA [5] is a Vietnamese multi-hop QA dataset built from Vietnamese Wikipedia, annotated with reasoning type (bridge / comparison) and gold supporting sentences; crucially, it uses the same format as HotpotQA, so our code runs on Vietnamese essentially unchanged. We use the distractor setting for all four, giving 10-20 candidate passages per question, so no Wikipedia index is required and cost is per question rather than per corpus. Because MuSiQue withholds test answers, we report on development sets throughout, and all four datasets are evaluated under a single reader (Qwen2.5-7B at 4-bit) so the numbers compare like with like; Table 3 gives the statistics.

Dataset	Lang.	#Q	#psg
HotpotQA (distractor)	En	7,405	10
2WikiMultiHopQA	En	12,576	10
MuSiQue-Ans	En	2,417	20
VimQA	**Vi**	**1,003**	**10**

Table 3. Datasets (development-set statistics).

5 Method

Given a question q and document set D, we seek a compressed context P_{comp} with $|P_{\text{comp}}| \ll |D|$ such that the reader still answers correctly. IterCOMP is training-free: all parameters stay frozen. Each document is decomposed into sentence-level evidence segments $e_{i,j}$. Figure 1 traces one question through two iterations.

5.1 Relevant evidence filtering

The *semantic* signal is the inner product of dense representations from a text encoder $E(\cdot)$:

$$S_{\text{sem}}(q, e_{i,j}) = E(q)^\top E(e_{i,j}). \tag{3}$$

This can miss exact lexical cues. Following M3-Embedding [1], the importance weight of token t is

$$w_t = \text{ReLU}(\mathbf{w}_{\text{lex}}^\top E(t)), \tag{4}$$

where $\mathbf{w}_{\text{lex}}$ is a projection vector. The *lexical* score is a sparse inner product over co-occurring tokens:

$$S_{\text{lex}}(q, e_{i,j}) = \sum_{t \in q \cap e_{i,j}} w_t^q \cdot w_t^{e_{i,j}}. \tag{5}$$

The two are combined into a dual-aspect relevance score:

$$S_{\text{dual}} = \lambda\, S_{\text{sem}} + (1 - \lambda)\, S_{\text{lex}}. \tag{6}$$

Filtering uses a percentile threshold rather than a fixed top-k:

$$E_{\text{cand}} = \{e_{i,j} \mid S_{\text{dual}} \geq \text{Percentile}(S, k)\}. \tag{7}$$

The threshold therefore *adapts* to each question's score distribution rather than committing to a fixed budget.

5.2 Answerability judgment and follow-up generation

These two decision points are what make IterCOMP "reasoning-aware": the query evolves with the evidence state instead of staying fixed at q. Neither prompt is published in the original paper; ours are released in full, in Vietnamese and English, with the code.

Parameter	Value
Max iterations	5
λ	0.6
k (percentile)	90; 85 for 2Wiki
Encoder	bge-m3 (frozen)
Reader	Qwen2.5-7B-4bit

Table 4. Hyperparameters, following the original paper (which used a LLaMA-3-8B reader); VimQA, absent there, uses $k = 90$ as HotpotQA.

6 Experimental Setup

We adopt the original paper's hyperparameters (Table 4).

6.1 Baselines

Raw Document concatenates everything with no compression. Oracle keeps gold supporting sentences only - the upper bound for any document-level method, and a diagnostic: if Oracle is also low, the bottleneck is the reader rather than the compressor. LLMLingua-2 [9] is our main compression baseline, chosen because its token classifier is natively multilingual and covers Vietnamese. Against these we run IterCOMP as reimplemented here.

6.2 Metrics

We report Exact Match (EM) and F1 under the standard HotpotQA normalisation, plus the compression ratio (compressed over Raw Document tokens). That normalisation is built for English, so for Vietnamese we preserve diacritics - stripping them collapses distinct words (*má* / *mà*) and renders EM/F1 meaningless. The released code documents this and the other text-handling prerequisites.

7 Results and Discussion

7.1 Main results across four datasets

The central claim holds on all four. Sentence-level compression beats token-level LLMLingua-2 by +10.72 to +18.58 F1 (Table 5, last row), every paired bootstrap interval clear of zero. This is the paper's thesis, and the one result that does not depend on which dataset, reader or sample size we look at. It survives a wider field too: on VimQA ($n = 200$) IterCOMP leads all six baselines we added, but only LLMLingua-2 is multilingual and meets Vietnamese on equal terms, where the margin is +11.36 F1, CI $[+4.93, +17.84]$, $p = 0.0008$ against a 7.7 threshold. The other five carry an English-only component (GPT-2 scoring; NQ- and HotpotQA-trained RECOMP; English-fine-tuned CompAct) and appear in full in the released results.

	VimQA 1003	MuSiQue 1500	2Wiki 1500	HotpotQA 1500
Oracle	61.03	44.90	57.65	**71.40**
Raw	46.87	21.16	35.72	**59.24**
IterCOMP	**51.29**	**25.35**	**36.72**	54.58
LLML-2	35.31	14.63	19.24	36.00
−Raw	+4.42†	+4.19†	+1.00	**−4.66**†
−LLML-2	+15.98†	+10.72†	+17.48†	+18.58†

Table 5. Main results: F1*, Qwen2.5-7B at 4-bit; the second row gives n. The last two rows are IterCOMP minus that baseline. † paired bootstrap interval excludes zero. Bold marks whichever of Raw and IterCOMP is higher. Per-dataset discussion is in the released repository.

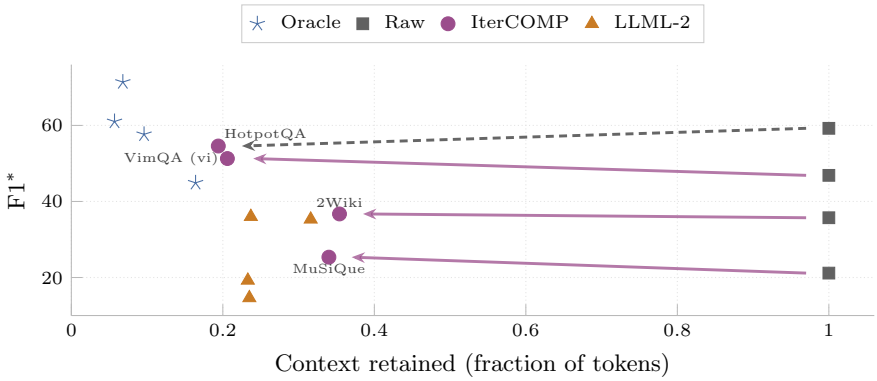


Fig. 3. Accuracy against context retained, all four datasets. Up and to the left is better. Each arrow runs from a dataset's Raw point to its IterCOMP point: on VimQA, MuSiQue and 2Wiki the arrow rises while moving left, so compression buys accuracy *and* tokens. On HotpotQA (dashed) it falls: the reader handles the full context well enough that filtering costs 4.66 F1, and only the token saving survives. Oracle marks the ceiling perfect selection would reach.

Method	MuSiQue		VimQA
	paper	ours	ours
Oracle	37.51	57.0	53.3
Raw Documents	19.92	32.8	43.8
LLMLingua-2	14.72	13.5	30.7
IterCOMP	**27.36**	**35.1**	**43.0**
ratio	*0.14*	*0.32*	*0.19*
Oracle/Raw	*1.88*	*1.74*	*1.22*

Table 6. F1 from the original paper (LLaMA-3-8B reader) against our reproduction (Qwen2.5-7B at 4-bit, $n = 50$), plus our Vietnamese results. VimQA has no published baseline; the paper does not evaluate it.

The full ordering does not. The paper reports Oracle > IterCOMP > Raw > LLMLingua-2. We reproduce it on VimQA (IterCOMP−Raw +4.42, interval clear) and on MuSiQue (+4.19, interval clear; at $n = 500$ the same margin read +3.31 and crossed zero, so tripling the sample decided it); the order matches on 2Wiki, where +1.00 does not separate; and on HotpotQA **Raw overtakes IterCOMP** by 4.66 F1, interval clear. The IterCOMP−Raw gap across the four is therefore +4.42, +4.19, +1.00 and −4.66 F1. Section 7.3 accounts for the HotpotQA reversal.

Small samples were inconclusive, not wrong. At $n = 50$ the IterCOMP-Raw gaps fell inside the paired detection threshold (18.0 F1) on both datasets, so we reported them as inconclusive and held the accuracy claim until the full 1,003-question VimQA run resolved it above; the underlying power analysis is released with the results. An early reading that the compression margin grows with task complexity likewise reverses at proper samples (+15.98 on VimQA at $n = 1{,}003$ against +10.72 on MuSiQue at $n = 1{,}500$), so we withdraw it.

7.2 Comparison with the original paper

Absolute numbers are not comparable across readers, but two invariants indicate a faithful reimplementation (Table 6). First, the **method ordering reproduces** - Oracle > IterCOMP > Raw > LLMLingua-2 on MuSiQue at both sample sizes, and on 2Wiki at scale (Section 7.1). Second, the Oracle/Raw ratio is in range: 1.74 at $n = 50$ and 2.12 at $n = 1{,}500$ against the paper's 1.88. That ratio is the headroom perfect selection buys over full context; being a property of the task and data loader rather than the reader, its agreement evidences that decomposition, gold-sentence labels and F1 are all correct.

Matching the operating point. A direct $n = 500$ comparison on MuSiQue puts IterCOMP 0.62 F1 *below* the paper's (26.74 against 27.36), while Oracle and Raw - which our compression code never touches - sit far above it (+12.33 and

+3.51), our stronger reader alone. That −0.62 is misleading: it compares systems at different operating points, 0.349 retention against 0.14. Sweeping k to match compression (released with the results) puts $k = 96$ at 0.160 retention and 29.04 F1, now +1.68 over the paper's 27.36. A residual gap remains - a reader +12.33 F1 stronger on perfect evidence should lift a working compressor by more than 1.68 - but it is within the noise floor, not the deficit the mismatched comparison implied.

7.3 The ordering is reader-dependent

Oracle exceeds Raw on every dataset while reading 5.7% to 14.4% of the tokens, confirming the premise the literature rests on: irrelevant passages *do* degrade accuracy. But the size of that gap decides whether compression can exploit it. Oracle/Raw is 2.13 on MuSiQue and 1.30 on VimQA, where IterCOMP holds its own, but only 1.21 on HotpotQA, where Raw wins outright. **The premise is reader-dependent**: a reader that already copes with distractors leaves compression little to recover. Swapping our 7B reader for Qwen2.5-0.5B on HotpotQA, the compression pipeline fixed, restores IterCOMP to +7.66 over Raw - reversing the −7.99 at 7B, and recovering the ordering the original paper reports with its own weaker reader. The HotpotQA reversal is thus a property of the reader, not a defect of the compressor.

7.4 Vietnamese-specific findings

Running the method on Vietnamese surfaces four results, developed in the appendices. Boolean answer normalisation is worth +10.0 F1 on Vietnamese ($p = 0.01$) and exactly 0.0 on English ($p = 1.00$), isolating a language-specific fix from a metric-inflating one. Nominal compression ratios understate token cost on Vietnamese by 2.39× because the tokeniser that bills is not the one that counts; we propose an *effective ratio* that divides fertility out. Vietnamese does not handicap the compressor: IterCOMP recovers 84% of the Oracle ceiling on VimQA.

Undiacriticised Vietnamese. Vietnamese is frequently typed without tone marks online, a setting no compression paper evaluates. Stripping diacritics costs 20.8 F1 as *má*, *mà* and *mã* collapse into one string; yet once the text is undiacriticised, **IterCOMP's compressed context beats even gold context** by 3.78 to 5.82 F1 (Table 7) - supplying precisely the gold sentences becomes *worse* than a compressed one. The released analysis traces the mechanism: stripping tone marks collapses 20.8% of VimQA's distinct syllables, and IterCOMP already beats Oracle on 9.8% of questions before any stripping, so the effect is amplified rather than created.

8 Error Analysis

We hand-categorised 50 IterCOMP predictions on VimQA (Table 8). Most errors belong to the *reader*: Oracle sees the gold evidence yet still reaches only 61.03

Setting	IterC.	Gold	Δ
`keep`	50.08	**54.94**	−4.86
`strip`	**29.29**	25.52	+3.78
`both`	**30.25**	24.43	+5.82

Table 7. Undiacriticised Vietnamese ($n = 200$, 7B reader). With tone marks (`keep`) gold context wins; strip them from the context (`strip`) or from context and question (`both`) and IterCOMP wins.

F1. Of IterCOMP's 48.7 F1 of error, 39.0 - **80%** - survives even perfect retrieval, so no compressor can address it.

Category	0.5B	7B
Fully correct (F1 = 1)	8%	16%
Partially correct (0 < F1 < 1)	34%	32%
Format error on yes/no	**32%**	**10%**
Fully wrong (F1 = 0)	26%	42%

Table 8. IterCOMP error distribution on VimQA ($n = 50$) under two readers. Format errors fall from 32% to 10% with the larger reader, a reader limitation rather than a compression failure.

With the 7B reader, **10%** of errors are not compression failures at all: yes/no questions answered correctly in substance but in the wrong surface form (F1 of 0). That bucket was **32%** under a 0.5B reader - "tức là yes" for "đúng", a small model code-switching into English - and falls on the identical prompt as the reader grows, so boolean answers need language-aware normalisation, released with the code, rather than a compression fix. Partially correct answers give a related but wrong entity within a correctly retained segment; fully wrong ones are mostly compositional, where the Raw-Oracle gap is widest, so distractors rather than the compressor drive them.

A further **9.8% of VimQA is a single question template** - "is X older than Y?" - on which 44.9% of answers score zero against 37.1% elsewhere, because the task is arithmetic rather than retrieval. Under a 0.5B reader the same template collapses to 92.1% zeros.

9 Limitations and Ethical Considerations

Limitations. Absolute numbers are not comparable to the paper's: no prompts were released (ours are released with the code), the reader differs (Qwen2.5-7B at 4-bit vs. LLaMA-3-8B), and our retention is 0.32 at $n = 50$ and 0.349 at $n = 500$ against their 0.14. Sample sizes are uneven but no longer thin: all four datasets

run at $n \geq 1{,}003$, the ablations at $n = 500$, the baselines at $n = 200$; what remains at $n = 50$ is the scorer ablation, reported as inconclusive. One baseline is missing, R2C releasing no public checkpoint; every other baseline with a released model is included, CompAct among them. The domain is narrow, since VimQA derives from Wikipedia. We do evaluate *undiacriticised* Vietnamese, with an unexpected result - see Table 7. Fertility is measured on translated text [2], so it holds for the tokenisers we bill against rather than the language at large. The sharpest limit is the gap between where fertility is measured and where compression is run: we report fertility for four writing systems but compress only Vietnamese and English, so the effective ratio is argued generally and demonstrated on one language pair. Hindi, the worst case at 4.97×, is where the correction would matter most and is exactly what we could not test - no multi-hop QA benchmark in Hindi or Thai carries the sentence-level supporting-fact annotations the Oracle branch needs.

Ethics. VimQA's CC BY-NC-SA 4.0 licence confines us to research use, and because compression is lossy - it can drop a negation with no loss of fluency, silently inverting an answer - sensitive deployments should verify outputs against the source; the fix worth 10.0 F1 in Vietnamese yet 0.0 in English is a further caution that "multilingual" models are not equally reliable across languages.

10 Conclusion

We reproduced IterCOMP without its code and gave it its first Vietnamese evaluation. The central claim holds - sentence-level beats token-level compression on all four datasets - but the full ordering is reader-dependent: a stronger reader gains enough from full context that compression's accuracy advantage narrows or reverses. Vietnamese itself does not handicap the compressor; what we could not move is the reader, failing on ∼30% of questions even given gold evidence.

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